

Karamjit Brar

Platform Engineer & AI Automation

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Technical Skills

Cloud & Platforms: AWS, GCP, Oracle Cloud, Kubernetes, EKS, Docker, VMware vSphere

Infrastructure & Automation: Terraform, Jenkins, GitLab CI/CD, Argo CD, Helm, Ansible, SaltStack

Observability & Monitoring: Datadog, Prometheus, Grafana

Programming & Scripting: Python, Bash, Groovy, SQL, Java, PowerShell

Operating Systems: RHEL, Ubuntu, Windows Server

Work Experience

System Engineer

Dec. 2023 – Present

Questrade, Toronto, ON

- Architected and Built a modular **VM provisioning pipeline** using Jenkins to deploy Virtual Machines on **vSphere and GCP**, reducing deployment time by 90% . Including guardrails with pre-provision checks (IP, DNS, AD) and post-provision integrations (Satellite, SaltStack, Device42).
- Engineered an AI-driven ChatOps agent using **LangChain**, enabling team interaction via Slack to automatically debug failed Jenkins builds, generate GitLab infrastructure MRs, and streamline cross-platform Jira workflows.
- Reduced DNS provisioning time from 2 days to minutes by engineering a **self-service DNS automation platform** that creates A and CNAME records on on-prem AD DNS servers through Jira requests, Slack approval gates, and Jenkins pipelines.
- Integrated Active Directory servers with **Datadog** by deploying agents, configuring integrations, and building customized pipelines in datadog to extract metrics from DNS logs.
- Automated AWS WorkSpaces lifecycle management by engineering a **serverless architecture (API Gateway, Lambda)** integrated with **Jira webhooks** to streamline on-demand provisioning.
- Reduced monthly cloud expenditures by **\$6,000** through an **AWS WorkSpaces** cost optimization solution; analyzed usage metrics to toggle 100+ low-usage contractor workspaces from monthly to hourly billing cycles.
- Deployed a high-availability **Prometheus** stack with Thanos sidecars and Grafana on a hybrid **EKS cluster** (AWS control plane, on-prem nodes), fully managed through Argo CD GitOps pipelines to enable scalable monitoring and logging.
- Engineered a Django-based self-service platform that enabled users to run networking and troubleshooting commands across distributed endpoints, using containerized execution on target VMs to provide safer connectivity.

Infrastructure Engineer Intern

May 2023 – Dec. 2023

Questrade, Toronto, ON

- Led a POC to transition VM management to automated SaltStack **highstates**, auditing states for idempotency, implementing custom Grains for dynamic targeting, and conducting knowledge-transfer sessions to align team practices.
- Built a **GitLab validation pipeline** for SaltStack, automatically testing Salt state changes against a QA environment prior to merge approval to block faulty configurations from reaching production.
- Hardened infrastructure security by migrating hard-coded credentials from code repositories to a **centralized Secret Server**, injecting them at runtime to eliminate plain-text exposure and enabled **automated password rotation**.
- Engineered a secure **AWS staging environment using Terraform** to deploy and evaluate a third-party acquisition, establishing strict security groups and ACLs to restrict data traffic exclusively to company devices for safe pre-production testing.

DevOps Engineer Intern

Sep. 2022 – Apr. 2023

CGI Inc., Markham, ON

- Led **migration and consolidation** of an outdated Jira Data Center instance to a **newer central instance**; developed a detailed step-by-step migration plan to ensure a seamless transition into the new environment.
- Managed both new and existing Jira projects, leading the transition from Excel to Jira to drive organization-wide adoption.

Education

Bachelor of Science, Computer Science — GPA: 3.8/4.0

York University, Toronto, ON

Projects

Kubernetes GitOps Homelab Platform
Oracle Cloud, Kubernetes, Argo CD, Helm, cert-manager, Terraform, Bitwarden, Istio, Prometheus, Grafana
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Personal Project

- Built a Kubernetes cluster on Oracle Cloud using Terraform and k3s, provisioning a 3-node cluster and cloud load balancer to host and manage containerized applications.
- Built a GitOps deployment workflow using GitHub and Argo CD ApplicationSet to automatically deploy and keep Kubernetes applications in sync with the Git repository.
- Integrated External Secrets with Bitwarden for secret management, cert-manager for TLS certificates, and Istio, Prometheus, and Grafana for service traffic monitoring and observability.

Certifications

Certified Kubernetes Administrator (CKA)	July 2025
AWS Certified Solutions Architect – Associate	Jan 2025
AWS Certified Developer – Associate	Sep 2024